

Assignment 15: Introduction to Plotly - P.1

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Part 1: Import libraries

```
import pandas as pd
import plotly.express as px
```

2. Loading dataset

```
url = "https://raw.githubusercontent.com/plotly/datasets/master/2011_us_ag_exports.csv"
export = pd.read_csv(url)

print(export.head())
```

	code	state	category	total exports	beef	pork	poultry	dairy	\
0	AL	Alabama	state	1390.63	34.4	10.6	481.0	4.06	
1	AK	Alaska	state	13.31	0.2	0.1	0.0	0.19	
2	AZ	Arizona	state	1463.17	71.3	17.9	0.0	105.48	
3	AR	Arkansas	state	3586.02	53.2	29.4	562.9	3.53	
4	CA	California	state	16472.88	228.7	11.1	225.4	929.95	

	fruits fresh	fruits proc	total fruits	veggies fresh	veggies proc	\
0	8.0	17.1	25.11	5.5	8.9	
1	0.0	0.0	0.00	0.6	1.0	
2	19.3	41.0	60.27	147.5	239.4	
3	2.2	4.7	6.88	4.4	7.1	
4	2791.8	5944.6	8736.40	803.2	1303.5	

	total veggies	corn	wheat	cotton
0	14.33	34.9	70.0	317.61
1	1.56	0.0	0.0	0.00
2	386.91	7.3	48.7	423.95
3	11.45	69.5	114.5	665.44
4	2106.79	34.6	249.3	1064.95

```
print(export.info())

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   code                   50 non-null    object
1   state                  50 non-null    object
2   category               50 non-null    object
3   total exports          50 non-null    float64
4   beef                   50 non-null    float64
5   pork                   50 non-null    float64
6   poultry                50 non-null    float64
7   dairy                  50 non-null    float64
8   fruits fresh           50 non-null    float64
9   fruits proc            50 non-null    float64
10  total fruits           50 non-null    float64
11  veggies fresh           50 non-null    float64
12  veggies proc            50 non-null    float64
13  total veggies           50 non-null    float64
14  corn                   50 non-null    float64
15  wheat                  50 non-null    float64
16  cotton                  50 non-null    float64
dtypes: float64(14), object(3)
memory usage: 6.8+ KB
None
```

Part 2: Visualization Tasks

1. Scatter plot:

- Create a scatter plot to visualize the relationship between total exports and beef.
- Parameters: "state" to color. Add title and axis labels.

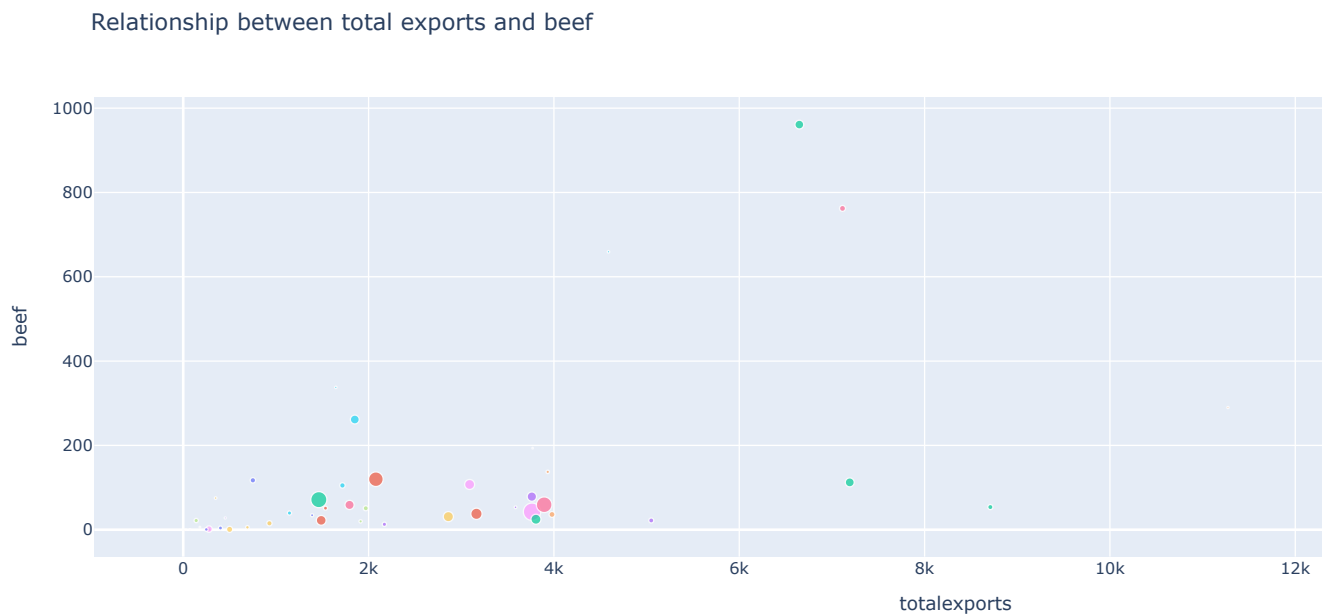
```
#remove spaces not need it and replace it
export.columns = export.columns.str.replace(" ", "_", regex=False)
#Check changes
print(export.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   code                   50 non-null    object  
1   state                  50 non-null    object  
2   category               50 non-null    object  
3   totalexports           50 non-null    float64 
4   beef                   50 non-null    float64 
5   pork                   50 non-null    float64 
6   poultry                50 non-null    float64 
7   dairy                  50 non-null    float64 
8   fruitsfresh            50 non-null    float64 
9   fruitsproc             50 non-null    float64 
10  totalfruits             50 non-null    float64 
11  veggiesfresh           50 non-null    float64 
12  veggiesproc            50 non-null    float64 
13  totalveggies           50 non-null    float64 
14  corn                   50 non-null    float64 
15  wheat                   50 non-null    float64 
16  cotton                  50 non-null    float64 
dtypes: float64(14), object(3)
memory usage: 6.8+ KB
None
```

```
print(export.columns.tolist())
```

```
['code', 'state', 'category', 'totalexports', 'beef', 'pork', 'poultry', 'dairy', 'fruitsfresh', 'fruitsproc', 'totalfruits', 'veggiesfresh',
```

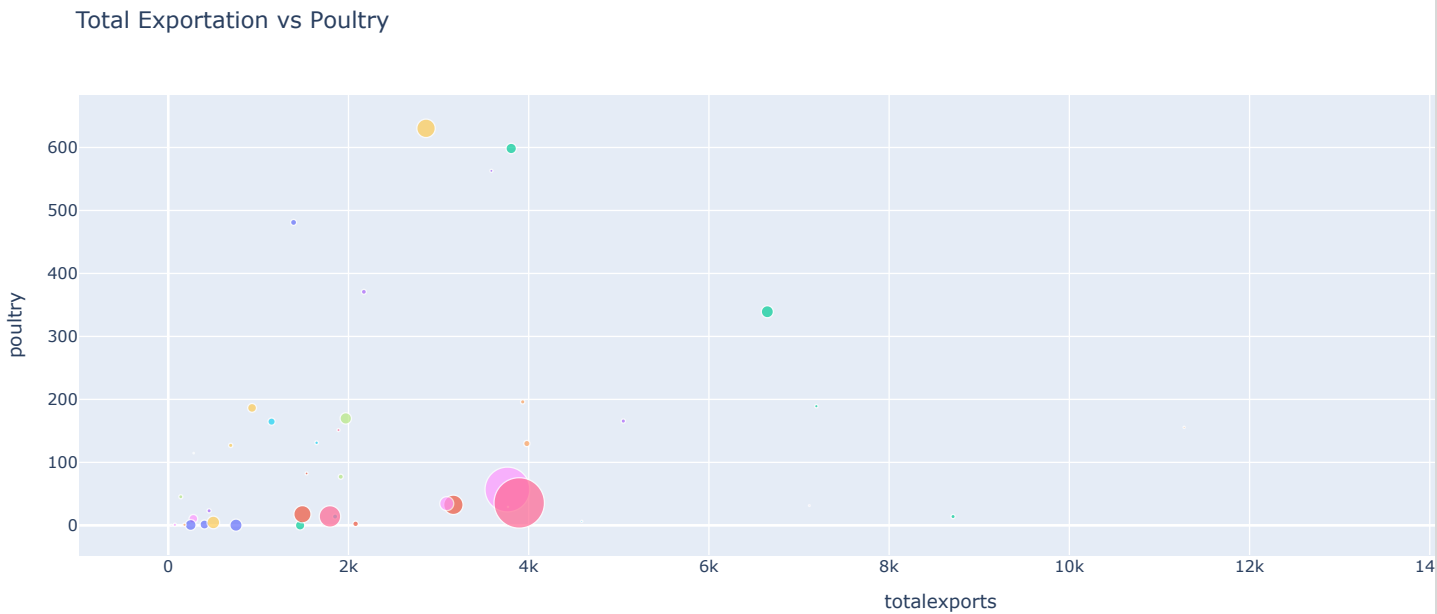
```
fig = px.scatter(export, x= 'totalexports', y='beef', color = 'state', size = 'totalveggies', title ="Relationship between total exports and beef")
fig.show()
```



2. Bubble plot:

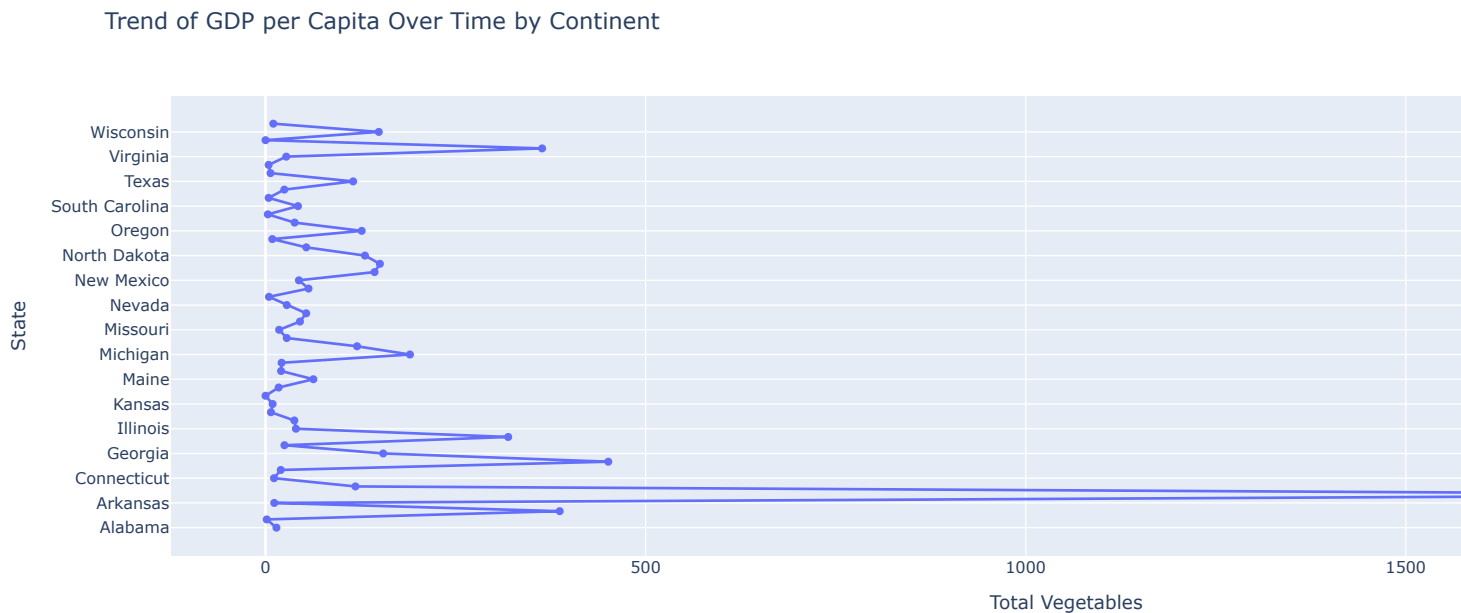
- Create a bubble chart to show total exports vs poultry with bubble sizes representing total fruits.
- parameters: "state" to color, 60 to size_max. Add title and axis labels.

```
fig = px.scatter(export, x = 'totalexports', y = 'poultry', size = 'totalfruits', color = 'state', size_max = 60, title = 'Total Exportation vs Poultry')
fig.show()
```



3. Line plot: Create an area plot for total veggies over the states. Add title and axis labels.

```
fig = px.line(export, x = 'totalveggies', y = 'state', markers = True, labels = {'totalveggies': 'Total Vegetables', 'state': 'State'}, title = 'Trend of GDP per Capita Over Time by Continent')
fig.show()
```

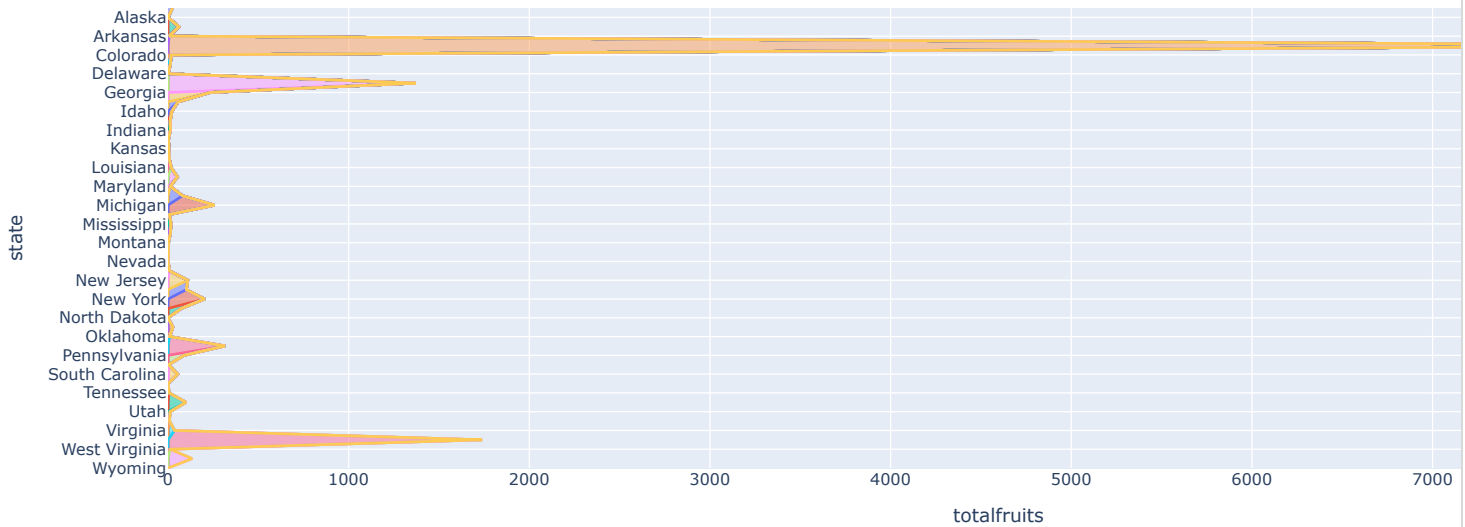


4. Area plot:

- Create an area plot for total fruits over the states. Add title and axis labels.

```
fig = px.area(export, x = 'totalfruits', y = 'state', color = 'state', title = 'Area Plot of GDP per Capita by Continent')
fig.show()
```

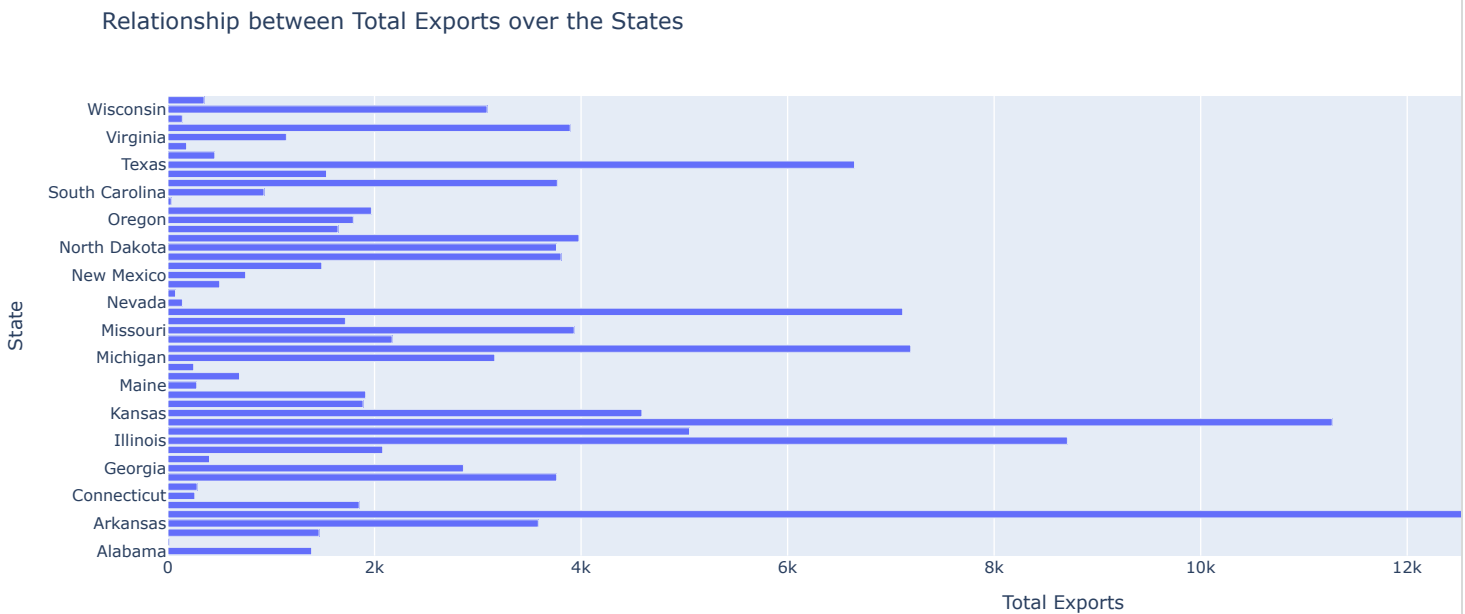
Area Plot of GDP per Capita by Continent



5. Bar plot:

- Create an area plot for total exports over the states. Add title and axis labels.

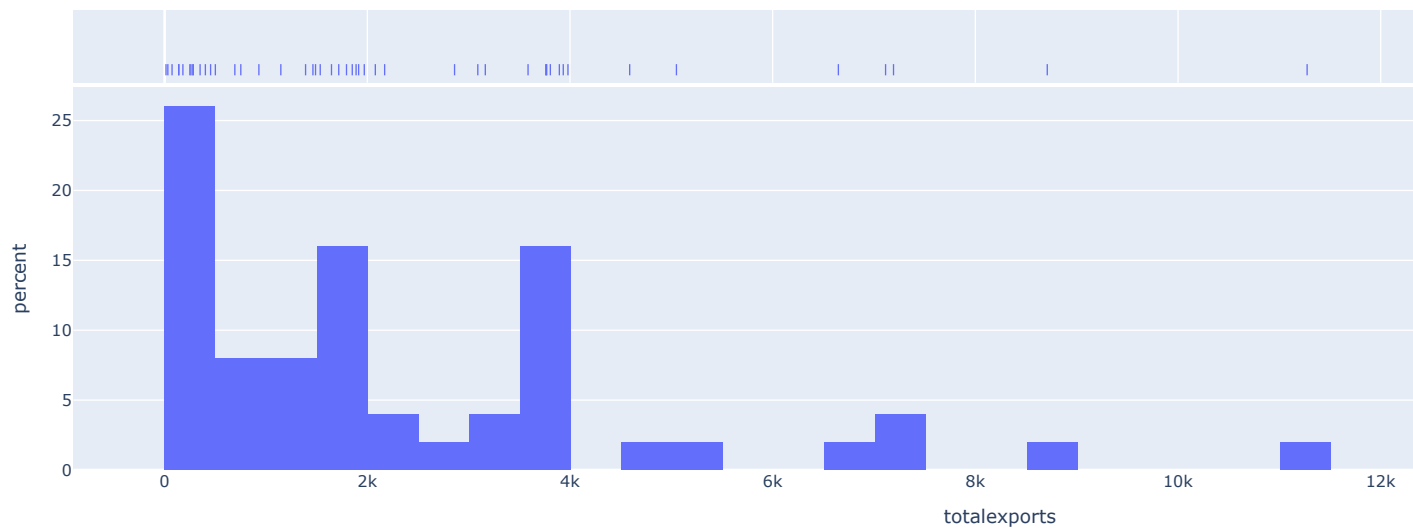
```
fig = px.bar(export, x='totalexports', y='state', title = 'Relationship between Total Exports over the States', labels = {'totalexports':'Total Exports'})
fig.show()
```



6. Histogram: Create a histogram for total exports. Add title and axis labels.

```
fig = px.histogram(export, x = 'totalexports', marginal = 'rug', histnorm = 'percent', nbins = 50, title = 'Distribution of Total Exports')
fig.show()
```

Distribution of Total Exports

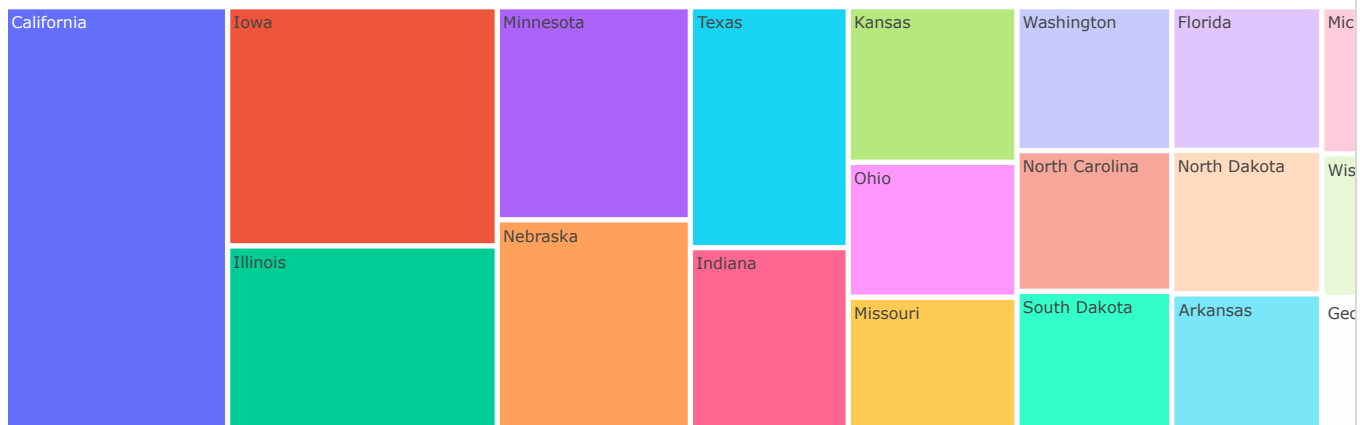


7. Tree map:

- Create a treemap plot to show the hierarchical relationship of state and total exports. Add title.

```
fig = px.treemap(export, path = ['state'], values = 'totalexports', title = 'The Hierarchical Relationship of State and Total Exports')
fig.show()
```

The Hierarchical Relationship of State and Total Exports

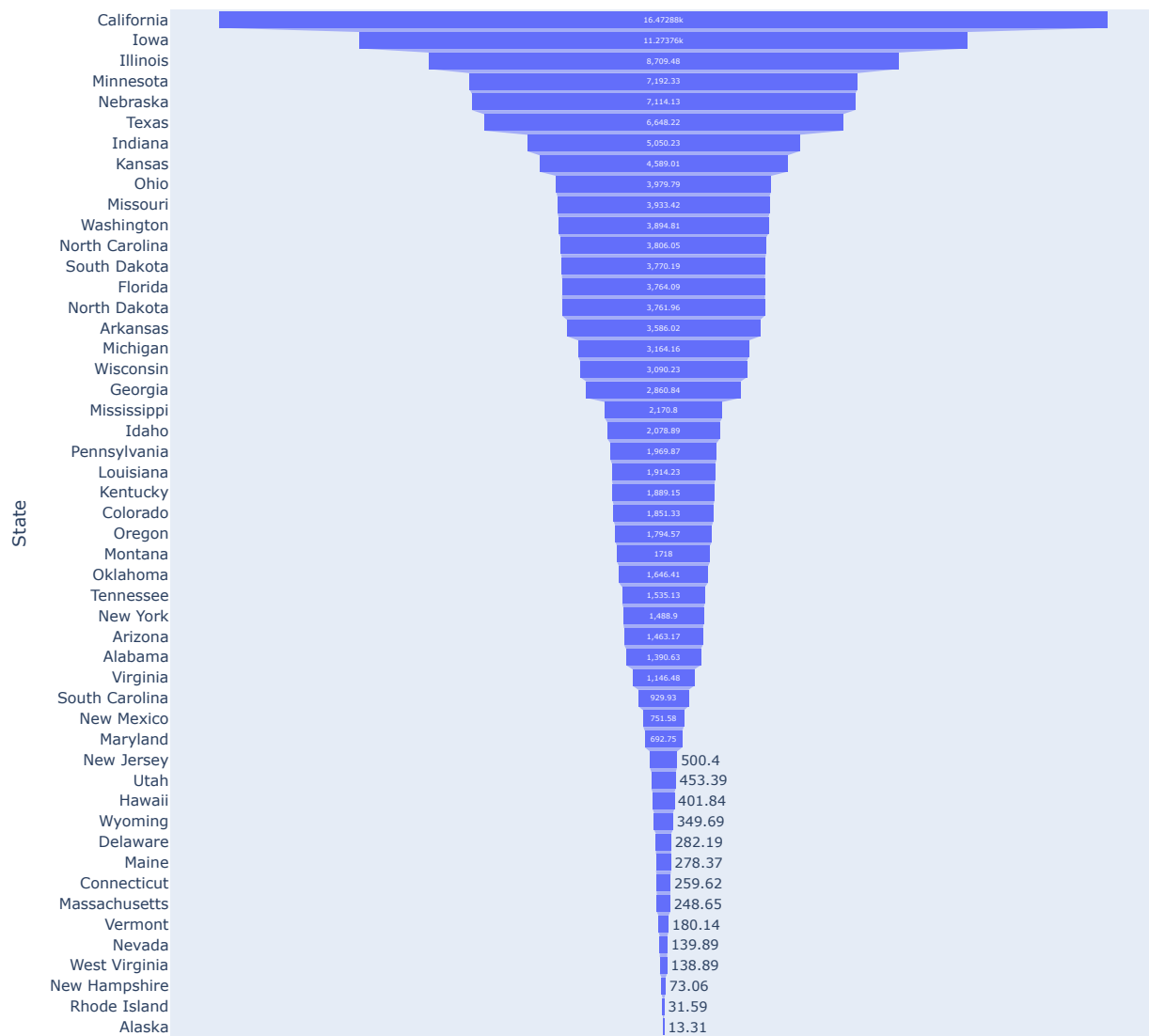


8. Funnel plot:

- Create a funnel plot for total exports by state.
- parameters: 1000 each to width and height. Add title and axis labels.

```
fig = px.funnel(export.sort_values('totalexports', ascending = False), x = 'totalexports', y = 'state',
                width = 1000, height = 1000, labels = {'totalexports': 'Total Exports', 'state': 'State'}, title = 'Population Comparison by Con
fig.show()
```

Population Comparison by Continent



9. Heatmap:

- Create a heatmap for the correlation of all numerical features/ columns. Add title.

```
corr = export.corr(numeric_only=True)
fig = px.imshow(corr, text_auto=True, aspect='auto', color_continuous_scale='RdBu_r', title='Correlation Heatmap of Numerical Features')
fig.show()
```

Correlation Heatmap of Numerical Features

totalexports	1	0.4587939	0.540014	0.2453099	0.5824717	0.6310866	0.6311103	0.6311009	0.6350839	0.6351072	0.6351072
beef	0.4587939	1	0.2045223	0.05279523	0.1794248	0.06955949	0.06956485	0.06956056	0.0838442	0.08386451	0.08386451
pork	0.540014	0.2045223	1	0.1877516	0.01060156	-0.07900709	-0.0789876	-0.07899041	-0.07825144	-0.07823678	-0.07823678
poultry	0.2453099	0.05279523	0.1877516	1	0.01161699	0.08363999	0.0836699	0.08365851	0.08121596	0.08121539	0.08121539
dairy	0.5824717	0.1794248	0.01060156	0.01161699	1	0.7209983	0.7209934	0.720999	0.7662058	0.7662192	0.7662192
fruitsfresh	0.6310866	0.06955949	-0.07900709	0.08363999	0.7209983	1	1	1	0.9696114	0.9696118	0.9696118
fruitsproc	0.631103	0.06956485	-0.0789876	0.0836699	0.7209934	1	1	1	0.9696121	0.9696125	0.9696125
totalfruits	0.6311009	0.06956056	-0.07899041	0.08365851	0.720999	1	1	1	0.9696128	0.9696132	0.9696132
veggiesfresh	0.6350839	0.0838442	-0.07825144	0.08121596	0.7662058	0.9696114	0.9696121	0.9696128	1	1	1
veggiesproc	0.6351072	0.08386451	-0.07823678	0.08121539	0.7662192	0.9696118	0.9696125	0.9696132	1	1	1
totalveggies	0.6351072	0.08386451	-0.07823678	0.08121539	0.7662192	0.9696118	0.9696125	0.9696132	1	1	1
corn	0.6311397	0.3254793	0.7700315	-0.01903263	0.04412922	-0.09182323	-0.09180837	-0.09180751	-0.09440644	-0.0943709	-0.0943709

```
fig = px.choropleth(export, locations='code', locationmode='USA-states', color='totalexports', scope='usa', hover_name='state', color_continuous_scale=fig.show())
```

Total Exports by State

